ServiceNow ITSM

Complete Workbook

Comprehensive Guide to IT Service Management
on the ServiceNow Platform

Covers: Incident | Problem | Change | Request | Knowledge
CMDB | SLA | Catalog | Discovery | ITOM | Administration
Workflows | Integrations | Reporting | Best Practices

Edition: 2026 | For Training & Certification Preparation

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Chapter 1: Introduction to ServiceNow & ITSM

1.1 What is ServiceNow?

ServiceNow is a cloud-based platform that provides enterprise service management capabilities. Originally focused on IT Service Management (ITSM), it has evolved into a multi-domain platform supporting HR, Customer Service, Security Operations, and custom applications. ServiceNow follows a single-instance, multi-tenant architecture delivered as Software-as-a-Service (SaaS).

Key Platform Characteristics

- Single data model: All applications share one underlying database and architecture.
- Configuration over customization: Prefer out-of-box features, configure via UI before coding.
- Now Platform: The foundation layer providing tables, forms, lists, workflows, and security.
- Scoped applications: ITSM, ITOM, CSM, HRSD are packaged as separate scoped apps.
- Instance-based: Each customer gets a dedicated logical instance (e.g., dev, test, prod).

1.2 What is ITSM?

IT Service Management (ITSM) refers to the activities, policies, and processes an organization uses to design, deliver, manage, and improve IT services aligned with business needs. ITSM is most commonly aligned with ITIL (Information Technology Infrastructure Library) best practices.

Core ITSM Processes in ServiceNow

Process	Purpose
Incident Management	Restore normal service operation ASAP
Problem Management	Identify and eliminate root causes of incidents
Change Management	Control and track changes to IT infrastructure
Request Management	Fulfill standard user service requests
Knowledge Management	Capture, share, and reuse organizational knowledge
Service Level Management	Define, monitor, and report on service quality
Configuration Management	Maintain accurate CI and relationship data (CMDB)

1.3 ITIL 4 vs ServiceNow

ITIL 4 introduces the Service Value System (SVS) and four dimensions of service management. ServiceNow ITSM applications map closely to ITIL practices but use ServiceNow-specific terminology and workflows. Understanding both frameworks helps during implementation and certification exams.

NOTE

ServiceNow certifications (CSA, CIS-ITSM, CAD) test practical platform knowledge. This workbook aligns content with common exam topics while focusing on real-world application.

EXERCISE: Chapter 1 Review

1. Log into a ServiceNow Personal Developer Instance (PDI) at developer.servicenow.com.
2. Navigate the Application Navigator and identify ITSM applications.
3. List five ITSM processes and describe the purpose of each in your own words.
4. Explore the ServiceNow Store and identify three ITSM-related scoped applications.

Chapter 2: Platform Architecture & Navigation

2.1 Instance Architecture

A ServiceNow instance consists of databases, application servers, and a web tier. Data is stored in tables (similar to database tables), and the platform provides a metadata-driven UI layer that renders forms, lists, and related lists dynamically.

Core Platform Concepts

- Tables: Store records (e.g., incident, change_request, cmdb_ci). Each table extends a parent table.
- Records: Individual rows in a table, identified by sys_id (32-character GUID).
- Fields/Columns: Attributes on a table (short text, reference, choice list, etc.).
- Forms: UI for viewing/editing a single record.
- Lists: Tabular views of multiple records with filtering and sorting.
- Related Lists: Child records linked via reference fields on the parent form.
- Modules: Navigation entries in the Application Navigator pointing to lists, forms, or URLs.
- Applications: Logical groupings of modules (e.g., Incident Management under ITSM).

2.2 Table Inheritance & the Task Table

ServiceNow uses table extension (class hierarchy). The task table is the base for most ITSM process tables. When you extend a table, child tables inherit all fields from parent tables.

Key Task Table Extensions

Table	Label	Key Additional Fields
incident	Incident	caller_id, category, subcategory, impact, urgency
problem	Problem	problem_state, known_error, workaround
change_request	Change Request	type, risk, approval, cab_date
sc_task	Catalog Task	request_item, sc_catalog
sc_req_item	Requested Item	cat_item, quantity, price

2.3 Navigation Essentials

- Application Navigator (left panel): Browse applications and modules.
- Filter Navigator: Type to search modules, records, or favorites.
- Favorites: Star modules for quick access.
- History: Recently viewed records (click clock icon).
- UI16 vs Next Experience: Classic UI vs modern Workspace/Next Experience UI.
- List controls: Filter, group, sort, export, and personalize list layouts.

2.4 Personal Developer Instance (PDI)

ServiceNow provides free Personal Developer Instances for learning. PDIs include ITSM, ITOM, and other applications. Instances hibernate after inactivity and reset periodically. Register at developer.servicenow.com to obtain yours.

TIP

Always practice in a PDI or sub-production instance. Never experiment with workflows, scripts, or ACL changes directly in production without change control.

Chapter 3: Incident Management

3.1 Purpose & Scope

Incident Management restores normal service operation as quickly as possible and minimizes business impact. An incident is an unplanned interruption or reduction in quality of an IT service.

Incident Lifecycle States

State	Description	Typical Actions
1 - New	Incident logged, not yet assigned	Auto-assignment rules may fire
2 - In Progress	Actively being worked	Diagnosis, communication, escalation
3 - On Hold	Waiting on caller or vendor	Set on-hold reason, set follow-up
6 - Resolved	Fix applied, pending caller confirmation	Resolution code & notes required
7 - Closed	Confirmed resolved or auto-closed	Survey may be sent, SLA stops
8 - Canceled	Duplicate or invalid	Link to parent incident

3.2 Priority Matrix

Priority is typically calculated from Impact and Urgency using a lookup matrix (data lookup: dl_u_priority). Impact = effect on business; Urgency = time sensitivity.

Impact & Urgency Values

- 1 - High: Critical business function affected / immediate action needed
- 2 - Medium: Degraded service / action needed soon
- 3 - Low: Minor inconvenience / can wait

3.3 Assignment & Routing

Incidents are assigned to assignment groups and individual assignees. Assignment rules, routing configurations, and predictive intelligence can automate routing based on category, CI, location, or language.

Key Assignment Concepts

- Assignment Group: Team responsible (e.g., Network, Service Desk Tier 1).
- Assignee: Individual within the group working the ticket.
- Skills-based routing: Match incidents to agents with required skills.
- On-call schedules: Escalate to on-call engineer based on rota.
- Work queues: Agent Workspace unified queue for incidents, cases, etc.

3.4 Major Incident Management

Major Incident Management handles high-impact incidents affecting many users or critical services. ServiceNow provides Major Incident Management application with dedicated workflows, war room coordination, stakeholder communication, and post-incident review.

Major Incident Process Steps

- Detection and declaration of major incident
- Formation of major incident team (MIT)
- Communication to stakeholders via email, SMS, status page
- Parallel technical investigation and workaround
- Post-incident review (PIR) and problem record creation

3.5 Incident Relationships

- Parent/Child: Group related incidents under one parent
- Problem: Link to underlying problem record
- Change: Link to change that caused or fixes incident
- Configuration Item: Affected CI in CMDB
- Knowledge Article: Attach relevant KB articles

3.6 Agent Workspace for Incidents

Agent Workspace (ITSM Workspace) provides a modern, unified interface for agents. Features include split-view record layout, related records panel, chat/collaboration, Knowledge sidebar, and embedded analytics.

EXERCISE: Incident Management Lab

1. Create an incident as an end user via the Service Portal or Employee Center.
2. As an agent, pick up the incident from the queue and set it to In Progress.
3. Set Impact=1, Urgency=1 and verify Priority becomes 1 (Critical).
4. Place the incident On Hold with reason 'Awaiting Caller'.
5. Resolve with resolution code 'Solved (Permanently)' and close the incident.
6. Create a second incident and relate it as a child of the first.

Chapter 4: Problem Management

4.1 Purpose & Scope

Problem Management identifies the root cause of incidents and prevents recurrence. Unlike incidents (restore service fast), problems focus on diagnosis and permanent resolution. Problems may be reactive (from incidents) or proactive (trend analysis).

Problem vs Incident

Aspect	Incident	Problem
Goal	Restore service quickly	Find and eliminate root cause
User impact	Service is broken now	May have no immediate impact
Timing	Immediate response	Investigation over time
Output	Resolution/workaround	Root cause, permanent fix, known error

4.2 Problem Lifecycle

- New: Problem identified and logged
- Assess: Initial assessment and categorization
- Root Cause Analysis: Investigation using techniques (5 Whys, Kepner-Tregoe, etc.)
- Fix in Progress: Permanent fix being developed/tested
- Resolved: Root cause addressed, fix deployed
- Closed: Verified no recurrence

4.3 Known Error Database (KEDB)

When root cause is identified but permanent fix is not yet available, the problem becomes a Known Error with a documented workaround. Known errors link to knowledge articles and help agents resolve related incidents faster.

4.4 Problem Tasks

Problem tasks break investigation into trackable work items (e.g., gather logs, test hypothesis). They follow the task table model and can be assigned to different teams.

EXERCISE: Problem Management Lab

1. Create a problem from a resolved incident using the 'Create Problem' related link.
2. Document root cause analysis notes in the problem record.
3. Mark the problem as a Known Error with a workaround description.
4. Create a problem task assigned to the Network team.
5. Link multiple incidents to the same problem and verify the relationship.

Chapter 5: Change Management

5.1 Purpose & Scope

Change Management ensures standardized methods and procedures for efficient handling of all changes to IT infrastructure. It balances the need for change with the risk of disruption.

Change Types

Type	Risk	Approval	Example
Standard	Pre-authorized, l	Pre-approved template	Password reset procedure
Normal	Variable risk	CAB or peer approval	Server OS patch
Emergency	High urgency	ECAB post-approval	Critical security hotfix

5.2 Change Lifecycle States

- -5 New: Change request created
- -4 Assess: Risk and impact assessment
- -3 Authorize: Awaiting approval
- -2 Scheduled: Approved, scheduled for implementation
- -1 Implement: Change being deployed
- 0 Review: Post-implementation review
- 3 Closed: Change complete
- 4 Canceled: Change canceled

5.3 Risk Assessment

Change Risk is calculated from Risk Impact and Risk Likelihood. The Change Management Risk Assessment module provides questionnaires to score changes objectively. High-risk changes require additional scrutiny and CAB review.

5.4 Change Advisory Board (CAB)

The CAB reviews normal changes, assesses risk, and approves or rejects them. Emergency CAB (ECAB) handles urgent changes. CAB meetings can be managed in ServiceNow with agenda items, attendees, and decision tracking.

Change Models (Modern Approach)

Change Models replace legacy workflows with Flow Designer-based models. Each model defines states, transitions, approvals, and automations. Out-of-box models include Normal, Standard, Emergency, and templates for cloud and application changes.

5.5 Change Tasks

- Implementation tasks: Steps to deploy the change
- Backout tasks: Rollback plan if change fails
- Test tasks: Validation after implementation
- Each task can have different assignment groups and schedules

EXERCISE: Change Management Lab

1. Create a Normal change request with planned start/end dates.
2. Complete the risk assessment questionnaire and observe risk calculation.
3. Submit for approval and approve as CAB manager.
4. Move through Implement and Review states to Closed.
5. Create a Standard change from a pre-approved template.
6. Create an Emergency change and note the expedited workflow.

Chapter 6: Service Catalog & Request Management

6.1 Service Catalog Overview

The Service Catalog provides a structured way for users to request IT services, hardware, software, and access. Catalog items are organized into categories and can include multi-step order guides, variable forms, and approval workflows.

Key Catalog Tables

Table	Purpose
sc_catalog	Catalog container (e.g., Service Catalog, HR Catalog)
sc_category	Grouping of catalog items
sc_cat_item	Individual catalog item / service offering
sc_cat_item_guide	Order guide (bundle of items)
item_option_new	Variables/questions on catalog items
sc_request	Header record for a user's request
sc_req_item	Individual line item (RITM) within a request
sc_task	Fulfillment tasks for requested items

6.2 Catalog Item Design

Variables (Item Options)

- Single Line Text, Multi Line Text, Select Box, CheckBox, Date, Reference, etc.
- Variable sets: Reusable groups of variables across items
- Catalog client scripts: Dynamic behavior on the catalog form
- UI policies: Show/hide/mandatory fields based on conditions

Fulfillment Methods

- Workflow/Flow: Automated fulfillment process
- Execution plan: Sequential or parallel task creation
- Direct delivery: No fulfillment (informational items)

6.3 Request Lifecycle

User submits request -> sc_request created -> sc_req_item(s) generated -> approvals fire -> fulfillment tasks created -> items delivered -> request closed.

6.4 Service Portal & Employee Center

Service Portal (legacy) and Employee Center (Next Experience) are self-service frontends. Users browse the catalog, submit requests, track status, and access knowledge without logging into the agent interface.

6.5 Record Producers

Record producers are catalog items that create records in backend tables (e.g., incident, change) instead of standard request items. They provide a user-friendly form for structured record creation.

EXERCISE: Service Catalog Lab

1. Browse the Service Catalog and order a standard item (e.g., laptop request).
2. Track the request through approval and fulfillment stages.
3. Create a custom catalog item with at least three variables.
4. Add an approval workflow for manager approval.
5. Create a record producer that generates an incident from a portal form.

Chapter 7: Knowledge Management

7.1 Purpose & Knowledge Base Structure

Knowledge Management captures, organizes, and shares information to enable self-service and improve agent efficiency. Articles are organized in Knowledge Bases and Categories.

Article Lifecycle

- Draft: Author writing content
- Review: Subject matter expert review
- Published: Available to target audience
- Retired/Outdated: No longer recommended

7.2 Article Types & Properties

- Knowledge Base: Top-level container with access controls
- Category: Sub-grouping within a KB
- Article template: Standard sections (Symptoms, Cause, Resolution)
- Effective dating: Publish/validity date ranges
- Versioning: Track article revisions
- Translation: Multi-language knowledge support

7.3 Knowledge-Centered Service (KCS)

KCS integrates knowledge creation into the workflow. Agents capture knowledge while solving issues. KCS practices include search before create, reuse, and flag/ fix. ServiceNow supports KCS with article linking, flagging, and quality metrics.

7.4 Knowledge in Agent Workspace

- Knowledge sidebar: Suggested articles based on incident text
- Attach article to incident for caller reference
- Create article from resolution notes
- Knowledge feedback and ratings

EXERCISE: Knowledge Management Lab

1. Create a knowledge article using the standard IT template.
2. Submit for review and publish to the IT knowledge base.
3. Search for the article from an incident and attach it.
4. Flag an article as outdated and create a revised version.

Chapter 8: Service Level Management (SLA)

8.1 SLA Fundamentals

Service Level Agreements define expected service quality. ServiceNow SLA engine tracks task SLA records against defined schedules and conditions, providing breach warnings and escalation.

Key SLA Components

- SLA Definition: Name, table, duration, schedule, conditions
- Task SLA: Running instance on a specific task record
- Schedule: Business hours calendar (excludes weekends/holidays)
- Start condition: When SLA timer begins
- Stop condition: When SLA timer stops (met/breached)
- Pause condition: When timer pauses (e.g., On Hold)

8.2 SLA Stages

Stage	Meaning
In Progress	SLA actively running
Paused	Timer paused per pause condition
Completed	Met target before breach
Breached	Target time exceeded
Cancelled	SLA no longer applicable

8.3 SLA Breakdowns & Reporting

SLA breakdown defines milestone percentages (e.g., 50%, 75%, 90%) for warnings. Business Rules and Notifications alert assignees and managers before breach. SLA dashboards show compliance percentages by team, priority, and category.

8.4 OLA & UC

- OLA (Operational Level Agreement): Internal team-to-team commitments
- UC (Underpinning Contract): External vendor/supplier agreements
- Both use the same SLA engine with different definitions

EXERCISE: SLA Lab

1. Review existing SLA definitions for incidents (e.g., P1 Response 15 min).
2. Create an incident and observe Task SLA record creation.
3. Put incident On Hold and verify SLA pauses.
4. Resolve before breach and confirm SLA stage = Completed.

Chapter 9: Configuration Management (CMDB)

9.1 CMDB Purpose

The Configuration Management Database (CMDB) stores Configuration Items (CIs) and their relationships. It provides a single source of truth for IT infrastructure, enabling impact analysis, change planning, and incident correlation.

CI Classes (cmdb_ci hierarchy)

- cmdb_ci: Base CI class
- cmdb_ci_computer: Servers, desktops, laptops
- cmdb_ci_server: Server-specific attributes
- cmdb_ci_network_adapter, cmdb_ci_ip_address: Network components
- cmdb_ci_appl: Application CIs
- cmdb_ci_service: Business services
- cmdb_ci_database, cmdb_ci_storage: Data infrastructure

9.2 Relationships

Relationships connect CIs (e.g., Runs on, Depends on, Contains, Connects to). Relationship rules define valid connections between CI classes. The CI Relationship Builder visualizes dependency maps.

9.3 CMDB Health & Data Quality

CMDB Health Dashboard KPIs

- Completeness: Required fields populated
- Correctness: Valid field values
- Compliance: Adherence to standards
- Duplicate CIs: Identification and remediation

9.4 Identification & Reconciliation Engine (IRE)

IRE prevents duplicate CIs by matching incoming data against existing records using identification rules and reconciliation rules. Data sources (Discovery, Import Sets, manual entry) feed into IRE for deduplication.

9.5 Service Definitions

Business services map technical CIs to business capabilities. Service owners define service offerings, SLAs, and supporting infrastructure through the service modeling process.

EXERCISE: CMDB Lab

1. Create a server CI manually with key attributes (name, IP, OS).
2. Create a business service and relate the server with 'Runs on'.
3. Use CI Relationship Builder to visualize dependencies.
4. Run CMDB Health dashboard and review completeness scores.

Chapter 10: Asset Management & Hardware Asset Management

10.1 Asset vs Configuration Item

Aspect	Asset	Configuration Item (CI)
Focus	Financial/lifecycle tracking	Technical/service configuration
Table	alm_asset	cmdb_ci (linked via asset field)
Ownership	Finance/procurement	IT operations
Lifecycle	Acquire, deploy, retire, dispose	Install, configure, change, decommission

10.2 Asset Lifecycle States

- In Stock -> In Transit -> In Use -> In Maintenance -> Retired -> Consumed
- Asset records track cost, PO number, warranty, lease, depreciation
- Stockrooms and transfer orders manage inventory movement

10.3 Software Asset Management (SAM)

SAM tracks software licenses, entitlements, and compliance. It integrates with Discovery to detect installed software and reconcile against purchased licenses. Key tables: sam_sw_install, sam_sw_product, alm_license.

10.4 Hardware Asset Management (HAM)

HAM extends asset management for hardware procurement, stockroom management, asset reservation, and disposal workflows. It integrates with the Service Catalog for hardware request fulfillment.

Chapter 11: Discovery & Service Mapping

11.1 ServiceNow Discovery

Discovery automatically identifies IP-enabled devices and software in your network, populating the CMDB without manual entry. It uses probes and sensors sent from MID Server to scan target devices.

Discovery Components

- MID Server: On-premise agent that runs probes within customer firewall
- Discovery Schedule: Defines IP ranges, credentials, and timing
- Probes: Scripts that query devices (SNMP, SSH, WMI, PowerShell)
- Sensors: Process probe results and create/update CIs
- Credentials: Stored securely for authenticated scanning

11.2 Discovery Phases

- 1. Port Scan: Identify live IPs and open ports
- 2. Classification: Determine device type
- 3. Identification: Match to CI class
- 4. Exploration: Deep scan for details (OS, apps, services)
- 5. CI Creation/Update: Populate CMDB via IRE

11.3 Service Mapping

Service Mapping builds dynamic maps of application services by identifying traffic patterns and dependencies. It uses entry points (load balancers, URLs) and discovery patterns to map multi-tier applications automatically.

11.4 Cloud Discovery

Cloud Discovery extends scanning to AWS, Azure, and GCP resources. Service accounts and cloud APIs populate cloud CIs (cmdb_ci_vm_instance, etc.).

Chapter 12: Release Management

12.1 Purpose

Release Management plans, schedules, and controls the rollout of releases (collections of changes) to production. It coordinates development, testing, and deployment across teams.

Release Record Structure

- Release phase: Design, Build, Test, Deploy, Review
- Release tasks: Work items assigned to teams
- Related changes: Changes bundled in the release
- Environment management: Dev, Test, Staging, Production

12.2 DevOps Integration

ServiceNow DevOps integrates with CI/CD tools (Jenkins, GitLab, Azure DevOps) to track pipeline executions, correlate deployments with changes, and automate change creation from pipeline events.

Chapter 13: Event Management & IT Operations

13.1 Event Management

Event Management correlates monitoring alerts from multiple sources (Nagios, SolarWinds, Datadog, etc.) into actionable incidents. It reduces alert noise through deduplication, correlation rules, and CI binding.

Alert Lifecycle

- Alert received via connector or email
- Binding to CI in CMDB
- Correlation into alert group
- Automation: Create incident or auto-resolve
- Remediation: Run remediation workflow

13.2 ITOM Visibility

IT Operations Management (ITOM) includes Discovery, Service Mapping, Event Management, Cloud Management, and Operational Intelligence. It provides visibility into infrastructure health and automates operational responses.

13.3 MetricBase & Operational Intelligence

MetricBase stores time-series metrics from monitoring tools. Operational Intelligence applies anomaly detection and forecasting to predict issues before they impact users.

Chapter 14: Administration: Users, Groups, Roles & ACLs

14.1 User Management

Users (sys_user table) represent people who access the instance. Each user has a user ID, roles, groups, departments, and preferences.

User Record Key Fields

- User ID: Login identifier
- Roles: Permissions assigned directly or via group
- Groups: Team membership for assignment and access
- Department, Location, Manager: Organizational context
- Active: Controls login ability

14.2 Groups & Roles

Groups (sys_user_group) organize users into teams. Roles (sys_user_role) define permissions. Assign roles to groups to grant permissions to all members. Common ITSM roles:

Role	Capability
itil	Basic ITIL user - create/read incidents, requests
itil_admin	Full ITSM administration
sn_incident_write	Create and update incidents
sn_change_write	Create and update changes
catalog_admin	Manage service catalog
admin	Full system administration (use sparingly)

14.3 Access Control Lists (ACLs)

ACLs enforce record-level and field-level security. Each ACL specifies operation (read, write, create, delete), table/field, and conditions/roles required. ACLs evaluate in order; deny takes precedence.

ACL Types

- Table ACL: Applies to entire table
- Field ACL: Applies to specific field
- Conditional ACL: Uses script or condition for dynamic evaluation

14.4 Data Policies & UI Policies

- Data Policy: Server-side mandatory/read-only enforcement
- UI Policy: Client-side show/hide/mandatory on forms

- Client Scripts: JavaScript on form load, change, submit

EXERCISE: Administration Lab

1. Create a new user and assign the itil role.
2. Create an assignment group and add the user as a member.
3. Review ACLs on the incident table (incident.acl.*).
4. Create a UI policy to hide 'Resolution notes' until state is Resolved.

Chapter 15: Workflows, Flow Designer & Automation

15.1 Workflow vs Flow Designer

Feature	Workflow (Legacy)	Flow Designer
Editor	Graphical Workflow Editor	Flow Designer (low-code)
Status	Legacy, still supported	Recommended for new automations
Use cases	Complex branching	Approvals, notifications, integrations
Subflows	Subflows	Reusable subflows

15.2 Flow Designer Components

- Trigger: Record created/updated, scheduled, inbound webhook
- Actions: Built-in operations (create record, update, notify, ask for approval)
- Conditions: If/else branching
- Subflows: Reusable flow segments
- Spokes: Integration actions (Slack, Teams, Jira, etc.)

15.3 Business Rules

Business Rules run server-side JavaScript on database operations (before, after, async, display). Use for field calculations, validations, and related record updates. Always consider performance impact on bulk operations.

15.4 Notification & Email

Notifications define email/SMS templates triggered by events. Use notification conditions and recipient lists (users, groups, fields). Email layouts support Jelly templating and dynamic content from triggering records.

Chapter 16: Reporting, Dashboards & Performance Analytics

16.1 Reporting

ServiceNow Reports visualize data from any table. Report types include bar, pie, line, list, pivot, heatmap, and SQL-based reports (admin only).

Report Design Steps

- 1. Select source table (e.g., incident)
- 2. Choose report type
- 3. Configure grouping, stacking, aggregation
- 4. Apply filters (query conditions)
- 5. Set roles and share with users/groups

16.2 Dashboards

Dashboards combine multiple reports and performance analytics widgets on one page. Role-based dashboards serve different audiences (executive, manager, agent). Homepages can be set per role or user.

16.3 Performance Analytics

Performance Analytics (PA) provides advanced analytics with indicators, breakdowns, targets, and time-series trending. It enables data-driven service improvement through KPI tracking and threshold alerting.

PA Key Concepts

- Indicator: Metric definition (e.g., Incident Resolution Time)
- Breakdown: Dimension for slicing data (by priority, team, category)
- Target: Desired KPI value
- Collection job: Scheduled data collection for indicators

Chapter 17: Integrations & Data Import

17.1 Integration Methods

Method	Use Case
REST API (Table API)	CRUD operations on records from external systems
Import Sets	Bulk data load from CSV, Excel, or JDBC
Transform Maps	Map import set rows to target tables
IntegrationHub Spokes	Pre-built connectors (Jira, Azure, SAP)
MID Server	On-premise connectivity for Discovery and integrations
Event Rules	Inbound events from monitoring tools
Outbound Web Services	SOAP/REST calls from ServiceNow to external APIs

17.2 REST API Essentials

The Table API (api/now/table/{tableName}) supports GET, POST, PUT, PATCH, DELETE. Authentication via Basic Auth, OAuth, or API keys. Use sysparm_query for encoded queries, sysparm_fields to limit returned fields, and sysparm_limit for pagination.

Example API Call

```
GET /api/now/table/incident?sysparm_query=active=true^priority=1
Authorization: Basic {base64(user:password)}
Accept: application/json
```

17.3 Import Sets & Transform Maps

Import Sets load data into staging tables (u_import_*). Transform Maps define field mapping, coalesce fields (match existing records), and run transform scripts. Use for migration, bulk updates, and scheduled data sync.

Chapter 18: Virtual Agent, Now Assist & Self-Service

18.1 Virtual Agent

Virtual Agent provides conversational AI for self-service. Users interact via chat to get answers, submit requests, and check ticket status. Topics define conversation flows with intent recognition and entity extraction.

Virtual Agent Components

- Topics: Conversation flows triggered by user phrases
- Entities: Data extracted from user input (date, email, choice)
- NLU model: Natural language understanding for intent matching
- Channels: Portal, Teams, Slack, mobile

18.2 Now Assist (Generative AI)

Now Assist brings generative AI to ITSM: auto-summarize incidents, suggest resolutions, generate knowledge articles, and assist agents with contextual recommendations. It integrates with the Now Platform AI capabilities.

18.3 Employee Center

Employee Center is the Next Experience self-service portal. It provides unified access to catalog, knowledge, cases, and Virtual Agent in a modern, mobile-friendly interface.

Chapter 19: ITIL Alignment & Best Practices

19.1 Implementation Best Practices

- Start with out-of-box processes; customize incrementally
- Define clear roles and responsibilities before go-live
- Establish data governance for CMDB and catalog
- Use sub-production instances (dev, test, UAT) for changes
- Document configuration decisions in implementation wiki
- Train agents and end users before launch
- Measure KPIs from day one with PA and dashboards

19.2 Common Pitfalls to Avoid

- Over-customizing before understanding OOB capabilities
- Skipping CMDB data quality governance
- Creating too many catalog items without lifecycle management
- Granting admin role broadly instead of scoped roles
- Ignoring performance: unindexed queries, heavy business rules
- Not testing workflows with edge cases and bulk scenarios

19.3 Continuous Improvement

Use Problem Management and PA trends for continuous improvement. Conduct regular CAB retrospectives, knowledge article reviews, and SLA compliance analysis. Align improvement initiatives with business outcomes.

Chapter 20: Certification Prep & Practice Questions

20.1 ServiceNow Certification Paths

Certification	Focus Area
CSA (Certified System Administrator)	Platform administration fundamentals
CIS-ITSM (Certified Implementation Specialist)	ITSM application implementation
CAD (Certified Application Developer)	Development and scripting
CIS-APM	Asset and Portfolio Management
CIS-EM	Event Management
CIS-SAM	Software Asset Management

20.2 Practice Questions

Question 1

Q: Which table do Incident, Problem, and Change Request extend?
A) sys_task B) task C) sc_task D) wm_task
Answer: B) task

Question 2

Q: What happens to an SLA when an incident moves to 'On Hold' state?
A) SLA stops permanently B) SLA is cancelled C) SLA pauses D) SLA continues
Answer: C) SLA pauses (if pause condition is configured)

Question 3

Q: Which component runs probes within the customer's network for Discovery?
A) IntegrationHub B) MID Server C) Event Management D) Flow Designer
Answer: B) MID Server

Question 4

Q: What is the difference between a Standard and Normal change?
A: Standard changes are pre-authorized with low risk; Normal changes require assessment and approval.

Question 5

Q: Which record type does a Record Producer create?
A: Always sc_req_item B: Any configured target table C: Only incidents D: Only tasks
Answer: B) Any configured target table

Question 6

Q: What does IRE stand for in CMDB context?
A: Identification and Reconciliation Engine

Question 7

Q: Which ServiceNow tool is recommended for new automations?
A: Workflow Editor B) Flow Designer C) Business Rules D) Client Scripts
Answer: B) Flow Designer

Question 8

Q: Priority on an incident is typically calculated from:
A: Impact + Urgency B) Impact x Urgency matrix C) Category + Subcategory D) Assignment group
Answer: B) Impact x Urgency matrix (lookup)

EXERCISE: Final Capstone Project

1. Design an end-to-end scenario: User reports email outage via portal.
2. Agent creates incident, links CI, attaches KB article, escalates priority.
3. Major incident declared; problem record created after resolution.
4. Change request raised for permanent fix; CAB approves and implements.
5. Knowledge article published; SLA compliance measured on dashboard.
6. Document each step with screenshots from your PDI.

Appendix C: Step-by-Step Configuration Guides

C.1 Creating an SLA Definition

Navigate to SLA > SLA Definitions > New. Set Name (e.g., P1 Response - 15 min), Table: incident, Duration: 00:15:00, Schedule: 8-5 weekdays. Set Start condition: Priority changes to 1. Stop condition: State is Resolved or Closed. Pause condition: State is On Hold. Save and test with a P1 incident.

C.2 Creating a Catalog Item

Go to Service Catalog > Catalog Definitions > Maintain Items > New. Enter Name, Category, Catalogs (Service Catalog), and Price if applicable. In the Variables related list, add questions (e.g., Laptop Model as Select Box). Set Workflow or Flow for fulfillment. Under Availability, set roles/groups who can see the item. Publish when ready.

C.3 Setting Up Assignment Rules

Navigate to System Policy > Rules > Assignment. Create new rule on incident table. Set Order (lower = higher priority), Condition (e.g., Category is Network), Assign to Group: Network Support. Multiple rules evaluate top-down; first match wins.

C.4 Creating a Change Model

Open Change > Administration > Change Models > New. Define model name, table (change_request), initial state, and state transitions. Add approval activities for Authorize state. Configure conditions for auto-transition. Activate model and set as default if needed.

C.5 CMDB Identification Rules

Go to CMDB > Identification/Reconciliation > Identification Rules. Select CI class (e.g., cmdb_ci_linux_server). Define identifier entries (serial number, name+IP). Create reconciliation rules to decide which data source wins for each field on update.

C.6 Building a Flow in Flow Designer

Open Flow Designer > New Flow. Set name and application scope. Choose trigger: Record Created on incident. Add action: Update Record (set assignment group). Add Ask for Approval action for manager approval on high priority. Add Send Notification action. Activate flow and test.

C.7 Creating a Report and Dashboard

Reports > Create New. Select Type: Bar chart, Source: incident, Group by: Priority, Aggregation: Count. Save as 'Incidents by Priority'. Dashboards > New Dashboard, add the report widget. Assign dashboard to itil role

homepage.

C.8 REST API Testing with curl

```
curl -X GET 'https://INSTANCE.service-now.com/api/now/table/incident?sysparm_limit=5' \
  -H 'Accept: application/json' \
  -u 'admin:password'
```

Replace INSTANCE with your PDI name. Use POST to create records, PATCH to update. Always use HTTPS and service accounts with minimal required roles in production.

C.9 Import Set Data Load

System Import Sets > Load Data > Choose File (CSV). Select target import set table or create new. Map columns. Run transform map to target table (e.g., cmdb_ci). Review import set run logs for errors. Schedule recurring imports for ongoing sync.

C.10 Virtual Agent Topic Creation

Virtual Agent > Designer > New Topic. Define trigger phrases ('reset password', 'password help'). Build conversation flow: greeting, collect email entity, call password reset subflow, confirm completion. Publish topic and test in Virtual Agent simulator.

Appendix D: Additional Practice Questions (20)

Q9

Which field on incident determines who receives survey?
Answer: Caller (caller_id)

Q10

What table stores approval records?
Answer: sysapproval_approver

Q11

Coalesce field in transform map is used for:
Answer: Matching/updating existing records

Q12

Default incident priority when Impact=2, Urgency=2:
Answer: 3 (Moderate) - verify matrix

Q13

Which role is minimum to create incidents?
Answer: itil or sn_incident_write

Q14

sc_req_item is child of which table?
Answer: sc_request

Q15

Business Rule 'before' runs:
Answer: Before database operation commits

Q16

CMDB class for Windows servers:
Answer: cmdb_ci_win_server

Q17

Discovery requires what on-premise component?
Answer: MID Server

Q18

Knowledge article state for public visibility:

Answer: Published

Q19

Change type with pre-approval:
Answer: Standard

Q20

Flow Designer trigger for scheduled job:
Answer: Scheduled - recurring or once

Q21

Encoded query 'active=true^priority=1' means:
Answer: AND condition (active AND priority 1)

Q22

sys_id format:
Answer: 32-character hexadecimal GUID

Q23

UI Policy runs on:
Answer: Client-side (browser)

Q24

Data Policy runs on:
Answer: Server-side (always enforced)

Q25

Performance Analytics indicator collects data via:
Answer: Scheduled collection job

Q26

Record producer creates records in:
Answer: Configured target table

Q27

On Hold reason required when:
Answer: Setting incident to On Hold state

Q28

CAB approves which change type?
Answer: Normal (and some Emergency post-facto)

Appendix E: Key Field Reference by Process

E.1 Incident Key Fields

Field	Type	Description
number	Auto	INC0010001 format ticket number
caller_id	Reference	User who reported (sys_user)
category	Choice	High-level classification
subcategory	Choice	Detailed classification
impact	Choice	1=High, 2=Medium, 3=Low
urgency	Choice	1=High, 2=Medium, 3=Low
priority	Choice	Calculated from impact x urgency
assignment_group	Reference	Team owning the ticket
assigned_to	Reference	Individual agent
cmdb_ci	Reference	Affected configuration item
short_description	String	Brief summary (required)
description	Text	Detailed description
work_notes	Journal	Internal agent notes
comments	Journal	Customer-visible comments
resolution_code	Choice	Required on resolve
close_code	Choice	Required on close

E.2 Change Request Key Fields

Field	Type	Description
type	Choice	Standard, Normal, Emergency
risk	Choice	Calculated risk level
risk_impact	Choice	Impact if change fails
implementation_plan	Text	Steps to implement
backout_plan	Text	Rollback procedure
test_plan	Text	Validation steps
justification	Text	Business reason for change
cab_date	Date	Scheduled CAB meeting
start_date	Date/Time	Planned start
end_date	Date/Time	Planned end

E.3 Encoded Query Operators

Operator	Meaning	Example
=	Equals	priority=1
!=	Not equal	state!=7

>	Greater than	priority>2
IN	In list	priorityIN1,2
LIKE	Contains	short_descriptionLIKEemail
STARTSWITH	Begins with	numberSTARTSWITHINC
ISEMPTY	Is empty	assigned_toISEMPTY
^	AND	active=true^priority=1
^OR	OR	priority=1^ORpriority=2
^ORDERBY	Sort	^ORDERBYDESCopened_at

Appendix A: Glossary of Key Terms

ACL:
Access Control List - security rule for table/field access

CAB:
Change Advisory Board - approves change requests

CI:
Configuration Item - component in CMDB

CMDB:
Configuration Management Database

ECAB:
Emergency Change Advisory Board

IRE:
Identification and Reconciliation Engine

ITIL:
Information Technology Infrastructure Library

ITOM:
IT Operations Management

ITSM:
IT Service Management

KCS:
Knowledge-Centered Service

KEDB:
Known Error Database

MID Server:
Management, Instrumentation, and Discovery Server

MIT:
Major Incident Team

NLU:
Natural Language Understanding

OLA:
Operational Level Agreement

PA:
Performance Analytics

PDI:
Personal Developer Instance

PIR:
Post-Incident Review

RITM:
Requested Item (sc_req_item)

SAM:
Software Asset Management

SLA:
Service Level Agreement

SVS:

Service Value System (ITIL 4)

UC:

Underpinning Contract

Appendix B: Useful Tables & Reference Data

Common ITSM Tables

Table Name	Label	Description
incident	Incident	Service disruption records
problem	Problem	Root cause investigation
change_request	Change Request	Controlled changes
sc_request	Request	Catalog request header
sc_req_item	Requested Item	Catalog line item
sc_task	Catalog Task	Fulfillment task
task_sla	Task SLA	SLA instance on task
cmdb_ci	Configuration Item	Base CI table
kb_knowledge	Knowledge	Knowledge articles
sys_user	User	Platform users
sys_user_group	Group	User groups/teams
sysapproval_approver	Approver	Approval records

Useful Navigation Paths (Classic UI)

- Incidents: Self-Service > Incident > Create New (or Incident > Create New)
- Changes: Change > Create New
- Catalog Admin: Service Catalog > Catalog Definitions > Maintain Items
- SLA Definitions: SLA > SLA Definitions
- CMDB: Configuration > CI Class Manager
- Reports: Reports > View/Run
- Users: User Administration > Users
- Roles: User Administration > Roles

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